HW02 · Ch 22 Electric Field — Worked Solutions

Source: CST homework02 (Chapter 22), mxhuang@scut.edu.cn · study-slides (vetame21.github.io/study-slides)

1 An electric field is most directly related to:

A) the momentum of a test charge
B) the kinetic energy of a test charge
C) the potential energy of a test charge
D) the force acting on a test charge ✓ (answer)
E) the charge carried by a test charge

Answer: D

SOLUTION

By definition $\vec{E} = \vec{F}/q_0$: the field is the force per unit charge on a test charge.

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2 The diagram shows the electric field lines due to two charged parallel metal plates. We conclude that:

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A) the upper plate is positive and the lower plate is negative
B) a proton at X would experience the same force if it were placed at Y ✓ (answer)
C) a proton at X experiences a greater force than if it were placed at Z
D) a proton at X experiences less force than if it were placed at Z
E) an electron at X could have its weight balanced by the electrical force

Answer: B

SOLUTION

Between parallel plates the field lines are straight, parallel and evenly spaced, so the field is uniform ($E = \sigma/\varepsilon_0$). A proton feels the same force qE at X and at Y.

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3 An isolated charged point particle produces an electric field with magnitude E at a point 2 m away. A point at which the field magnitude is $E/4$ is:

A) 0.5 m away
B) 1 m away
C) 2 m away
D) 4 m away ✓ (answer)
E) 8 m away

Answer: D

SOLUTION

$E \propto 1/r^2$. To divide E by 4, multiply r by $\sqrt{4} = 2$: from 2 m to 4 m.

4 **An isolated charged point particle produces an electric field with magnitude E at a point 2 m away. At a point 1 m from the particle the magnitude of the field is:**

A) E
B) $2E$
C) $4E$ ✓ (answer)
D) $E/2$
E) $E/4$

Answer: C

SOLUTION

Halving r multiplies $1/r^2$ by 4, so the field is $4E$.

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5 **Two point particles with charges q_1 and q_2 are a distance r apart. The electric field is zero at a point P between them on the segment joining them. We conclude that:**

A) q_1 and q_2 must have the same magnitude and sign
B) P must be midway between the particles
C) q_1 and q_2 must have the same sign but may have different magnitudes ✓ (answer)
D) q_1 and q_2 must have equal magnitudes and opposite signs
E) q_1 and q_2 must have opposite signs and may have different magnitudes

Answer: C

SOLUTION

Between the charges, the two fields can only point opposite ways if the charges have the SAME sign. The zero point then shifts toward the smaller charge, so the magnitudes can differ.

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6 **The diagrams depict four different charge distributions, all with particles the same distance from the origin. The electric field at the origin:**

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A) is least for situation 1
B) is greatest for situation 3
C) is zero for situation 4 ✓ (answer)
D) is downward for situation 1
E) is downward for situation 3

Answer: C

SOLUTION

In situation 4 the charges are arranged symmetrically so that their fields cancel in pairs at the origin.

7 **Two point particles, $+8 \times 10^{-9}$ C and -2×10^{-9} C, are 4 m apart. The electric field midway between them is:**

A) 9×10^{9} N/C
B) 13,500 N/C
C) 135,000 N/C
D) 36×10^{-9} N/C
E) 22.5 N/C ✓ (answer)

> **Answer: E**

SOLUTION

At the midpoint (2 m from each), both fields point from + toward –, so they ADD: $E = k\frac{8\times10^{-9}}{2^2} + k\frac{2\times10^{-9}}{2^2} = 18 + 4.5 = 22.5$ N/C.

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8 **Two point charges are arranged as shown. In which region could a third charge +1 C be placed so that the net electrostatic force on it is zero?**

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A) I only ✓ (answer)
B) I and II only
C) III only
D) I and III only
E) II only

> **Answer: A**

SOLUTION

For opposite charges, the zero point lies OUTSIDE the pair, on the side of the smaller-magnitude charge (here region I), where $k\frac{2}{r_1^2} = k\frac{4}{r_2^2}$ can balance.

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9 **Positive charge $+Q$ is uniformly distributed on the upper half of a semicircular rod and $-Q$ on the lower half. What is the direction of the electric field at P, the center of the semicircle?**

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> The answer arrows were images in the original file; official answer B.

> **Answer: B**

SOLUTION

Each half's field at P points away from $+Q$ and toward $-Q$, so the horizontal parts cancel and the total field points straight DOWN (from the + half toward the – half).

10 **Positive charge $+Q$ is uniformly distributed on the upper half of a rod and $-Q$ on the lower half. What is the direction of the electric field at P, on the perpendicular bisector of the rod?**

> Figure not available: the original homework file was missing when these solutions were made.

> The answer arrows were images in the original file; official answer B.

> **Answer: B**

SOLUTION

By the same symmetry as Q9, the components along the bisector cancel and the field points from the + half toward the – half, parallel to the rod (downward).

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11 **A disk with uniform positive surface charge density 2.5×10^{-6} C/m² lies in the xy plane, centered on the origin, with radius 7.5 cm. What is the electric field at $z = 15$ cm?**

A) 30 N/C
B) 300 N/C
C) 3000 N/C
D) 1.5×10^{4} N/C ✓ (answer)
E) 3.0×10^{7} N/C

> **Answer: D**

SOLUTION

Use the charged-disk result from Ch 22:

$$E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right), \qquad \frac{\sigma}{2\varepsilon_0} = \frac{2.5 \times 10^{-6}}{2(8.85 \times 10^{-12})} = 1.41 \times 10^{5}\ \mathrm{N/C}.$$

$$\sqrt{0.15^2 + 0.075^2} = 0.1677\ \mathrm{m} \ \Rightarrow\ E = 1.41 \times 10^{5}\,(1 - 0.894) = 1.5 \times 10^{4}\ \mathrm{N/C}.$$

Why this method: the point is on the disk's axis, which is exactly where the formula applies.

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12 **The electric field due to a uniform distribution of charge on a spherical shell is zero:**

A) everywhere
B) nowhere
C) only at the center of the shell
D) only inside the shell ✓ (answer)
E) only outside the shell

> **Answer: D**

SOLUTION

Gauss' law: a sphere inside the shell encloses no charge, so $E = 0$ everywhere inside. Outside, it equals kq/r^2.

13 **A charged oil drop with a mass of 2×10^{-4} kg is held suspended by a downward electric field of 300 N/C. The charge on the drop is:**

A) $+1.5 \times 10^{-6}$ C
B) -1.5×10^{-6} C
C) $+6.5 \times 10^{-6}$ C
D) -6.5×10^{-6} C ✓ (answer)
E) 0 C

Answer: D

SOLUTION

Balance: $|q|E = mg \Rightarrow |q| = \frac{(2\times10^{-4})(9.8)}{300} = 6.5 \times 10^{-6}$ C. The electric force must point UP while $\vec{E}$ points down, so q is negative.

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14 **A uniform electric field of 300 N/C makes an angle of 25° with the dipole moment of an electric dipole. If the torque has a magnitude of 2.5×10^{-7} N·m, the dipole moment must be:**

A) 8.3×10^{-10} C·m
B) 9.2×10^{-10} C·m
C) 2.0×10^{-9} C·m ✓ (answer)
D) 8.3×10^{-5} C·m
E) 1.8×10^{-4} C·m

Answer: C

SOLUTION

$\tau = pE\sin\theta \Rightarrow p = \frac{2.5\times10^{-7}}{300\ \sin 25^\circ} = 2.0 \times 10^{-9}$ C·m.

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15 **A dipole ($p = 2 \times 10^{-9}$ C·m) in a 300 N/C field is initially perpendicular to the field and rotates until it points along the field. The work done by the field is:**

A) -12×10^{-7} J
B) -6×10^{-7} J
C) 0 J
D) 6×10^{-7} J ✓ (answer)
E) 12×10^{-7} J

Answer: D

SOLUTION

$W = -\Delta U = -(U_f - U_i)$ with $U = -pE\cos\theta$: $U_i = 0$ (at 90°), $U_f = -pE$ (at 0°). So $W = pE = (2 \times 10^{-9})(300) = 6 \times 10^{-7}$ J.

16 **Charge Q is distributed uniformly along a semicircle of radius a. Which formula gives the magnitude of the force on a particle of charge q at the center of the circle?**

A) $\frac{1}{4\pi\varepsilon_0}\frac{qQ}{\pi a}$
B) $\frac{1}{4\pi\varepsilon_0}\frac{qQ}{\pi a^2}$
C) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{\pi a}$
D) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{\pi a^2}$ ✓ (answer)
E) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{a^2}$

Answer: D

SOLUTION

Same method as the 120° arc in Ch 22, but over 180°:

$$E = \frac{\lambda}{4\pi\varepsilon_0 a}\int_{-\pi/2}^{\pi/2}\cos\theta\, d\theta = \frac{2\lambda}{4\pi\varepsilon_0 a}, \qquad \lambda = \frac{Q}{\pi a}.$$

$$F = qE = \frac{1}{4\pi\varepsilon_0}\frac{2qQ}{\pi a^2}.$$

Why this method: by symmetry the perpendicular components cancel, so only $\int dE\cos\theta$ survives. A quick check: the units must be qQ/length^2, which rules out A and C.

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17 **A particle ($m = 20$ mg, $q = -5.0$ μC) moves in a uniform electric field of 60 N/C in the $+x$ direction. At $t = 0$ it moves at 30 m/s in the $+x$ direction through the origin. Determine the maximum distance beyond $x = 0$ it travels in the $+x$ direction.**

A) 25 m
B) 20 m
C) 15 m
D) 30 m ✓ (answer)
E) 60 m

Answer: D

SOLUTION

The charge is negative, so it is pushed back: $a = \frac{qE}{m} = \frac{-(5\times10^{-6})(60)}{2\times10^{-5}} = -15$ m/s². It stops when $0 - 30^2 = 2(-15)x$, so $x = 30$ m.

18 **A particle (mass 5.0 g, charge 40 mC) is in a uniform field $E_x = 2.5$ N/C, $E_y = E_z = 0$. At $t = 0$ its velocity is $v_y = 50$ m/s. What is its speed at $t = 2.0$ s?**

A) 81 m/s
B) 72 m/s
C) 64 m/s ✓ (answer)
D) 89 m/s
E) 25 m/s

Answer: C

SOLUTION

Acceleration only along x: $a_x = \frac{qE}{m} = \frac{(0.040)(2.5)}{0.005} = 20$ m/s², so $v_x = 40$ m/s at 2 s while v_y stays 50 m/s. Speed $= \sqrt{40^2 + 50^2} = 64$ m/s.

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19 **Which one of the diagrams is not a possible electric field configuration for a region of space that contains no charges?**

Figure not available: the original homework file was missing when these solutions were made.

The diagrams were images in the original file; official answer D (from the answer form).

Answer: D

SOLUTION

Electric field lines can never cross, and in a charge-free region they can't start or stop. The impossible diagram breaks one of these rules.

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20 **A 24 nC charge is distributed uniformly along the x axis from $x = 2$ m to $x = 6$ m. Which integral gives the magnitude (in N/C) of the electric field at $x = +8$ m?**

A) $\int_2^6 \frac{54\,dx}{(8-x)^2}$ ✓ (answer)
B) $\int_2^6 \frac{216\,dx}{(8-x)^2}$
C) $\int_2^6 \frac{54\,dx}{x^2}$
D) $\int_2^6 \frac{216\,dx}{x^2}$
E) none of these

Answer: A

SOLUTION

$\lambda = 24\ \text{nC}/4\ \text{m} = 6$ nC/m, so $k\lambda = (9 \times 10^9)(6 \times 10^{-9}) = 54$. A piece at x is $(8 - x)$ from the point: $dE = \frac{54\,dx}{(8-x)^2}$.

Problem 1. Two charged particles, 5.00 nC and –20.00 nC, are placed as shown in the figure. Find the electric field at the origin. ($k = 9.0 \times 10^9$ N·m²/C²)

> Figure not available: the original homework file was missing when these solutions were made.

> **Answer:** $\vec{E} = (1.3 \times 10^5\,\hat{\imath} + 4.0 \times 10^4\,\hat{\jmath})$ N/C

From the official solution: the 5.00 nC charge is 3.0 cm away and gives $E_+ = k\frac{5.00\times10^{-9}}{(0.030)^2} = 5.0 \times 10^4$ N/C along $+\hat{\imath}$. The –20.00 nC charge sits at $(0.040, 0.020)$ m and gives a field pointing TOWARD it:

$$E_- = k\frac{20.00 \times 10^{-9}}{0.040^2 + 0.020^2} = 9.0 \times 10^4\ \text{N/C} \ \Rightarrow\ (8.0 \times 10^4\,\hat{\imath} + 4.0 \times 10^4\,\hat{\jmath})\ \text{N/C}.$$

Sum: $\vec{E} = (1.3 \times 10^5\,\hat{\imath} + 4.0 \times 10^4\,\hat{\jmath})$ N/C.

Why this method: Superposition: each field from kq/r^2, its direction from the unit vector; add components.

Problem 2. Two point charges, –6.25 nC and –12.5 nC, are 25.0 cm apart (figure). Find the net field at (a) point A and (b) point B. (c) What force would act on a proton at A?

> Figure not available: the original homework file was missing when these solutions were made.

(a)

> **Answer:** 8.75×10^3 N/C, to the right

At A (0.150 m from q_1, 0.100 m from q_2) the fields point in opposite directions:

$$E_{A1} = k\frac{6.25 \times 10^{-9}}{0.150^2} = 2.50 \times 10^3, \qquad E_{A2} = k\frac{12.5 \times 10^{-9}}{0.100^2} = 1.125 \times 10^4\ \text{N/C}.$$

Net $= 1.125 \times 10^4 - 2.50 \times 10^3 = 8.75 \times 10^3$ N/C, to the right.

Why this method: Opposite directions → subtract magnitudes.

(b)

> **Answer:** 6.54×10^3 N/C, to the right

At B (0.100 m from q_1, 0.350 m from q_2) both fields point the same way:

$$5.625 \times 10^3 + 9.18 \times 10^2 = 6.54 \times 10^3\ \text{N/C}.$$

Why this method: Same direction → add magnitudes.

(c)

> **Answer:** 1.40×10^{-15} N, to the right

$F = qE = (1.60 \times 10^{-19})(8.75 \times 10^3) = 1.40 \times 10^{-15}$ N, along $\vec{E}$ because the proton is positive.

Why this method: $\vec{F} = q\vec{E}$.

Problem 3. Two particles with charges $q_1 = 0.500$ nC and $q_2 = 8.00$ nC are 1.20 m apart. At what point along the line joining them is the total electric field zero?

> **Answer:** 0.24 m from q_1 (0.96 m from q_2)

Like charges, so the zero point is BETWEEN them, closer to the smaller charge. With x measured from q_1:

$$\frac{0.500}{x^2} = \frac{8.00}{(1.20-x)^2} \;\Rightarrow\; (1.20-x)^2 = 16x^2 \;\Rightarrow\; 1.20 - x = 4x \;\Rightarrow\; x = 0.24\ \text{m}.$$

Why this method: Equal magnitudes, opposite directions; the square root turns it into a linear equation.

Problem 4. A positively charged rod of length $L = 0.250$ m with $\lambda = 2.33$ μC/m lies along the x axis from $x = 0$ to $x = L$. Find the electric field at P, a distance 0.375 m from the origin on the x axis.

> Figure not available: the original homework file was missing when these solutions were made.

> **Answer:** 1.12×10^5 N/C, along $+x$

Each piece $dq = \lambda\, dr$ at position r is $(x - r)$ from P:

$$E = \int_0^L \frac{k\lambda\, dr}{(x-r)^2} = k\lambda \left[\frac{1}{x-r}\right]_0^L = \frac{k\lambda L}{x(x-L)}.$$

$$E = \frac{(9.0\times10^9)(2.33\times10^{-6})(0.250)}{(0.375)(0.125)} = 1.12\times10^5\ \text{N/C}.$$

Why this method: Continuous charge: slice it, use the point-charge field for each slice, integrate. All slices point the same way, so no components are needed.